MILITARY SPECIFICATION

CONVERTER, TELEPHONE SIGNAL CV-1919( )/G

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of converter, designated as Converter, Telephone Signal CV-1919( )/G, for use as a three channel integration device between automatic 4-wire telephone central offices and manual 2-wire telephone switchboards.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-handling, Wood (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced
PPP-F-320	Fiberboard, Corrugated, and Solid Sheet Stock (Container Grade) and Cut Shape
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard

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SPECIFICATIONS (Cont'd)

FEDERAL

PPP-B-640	Box Fiberboard, Corrugated, Triple Wall
PPP-S-760	Strapping, Nonmetallic (and Connectors)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for packaging

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads (40 in x 48 in, 4-way Partial and 4-way Pallets)
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging

DRAWINGS

SIGNAL CORPS

DL-SM-B-743540	Converter, Telephone Signal CV-1919( )/G
SM-A-743617	Production Test Procedure for Converter, Telephone Signal CV-1919( )/G

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and Drawing and Data List DL-SM-B-743540 (Including the requirements for parts, materials and processes thereon).

3.3 Parts, Materials, and Processes: General. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein shall apply (see 4.4).

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (see 4.4).

3.5 Marking (See 4.4).

3.5.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial Numbers. The equipment shall be serially numbered.

3.6 Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the tests of 4.6. Each equipment on order shall be subjected to a burn-in of a minimum of 48 hours with the last 24 hours failure free (4.6.2). (See also 4.5).

3.7 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.7).

3.8 Electrical.

3.8.1 Operation.

3.8.1.1 Incoming Call. Each channel of the equipment shall accept incoming calls from 4-wire automatic switchboards. A seize signal of 2250 Hz shall be detected on the Receive pair of terminals and a seize acknowledge signal of 570 Hz ± 7.4 Hz at -14 dBm ± 2 dB shall be returned on the send pair of terminals, the service request indicator and night alarm shall be activated. Upon the termination of the incoming seize signal, the seize acknowledge signal shall be sent for an additional 250 ms. (milliseconds). After the seize acknowledge has stopped, a ring back signal of 425 Hz ± 5.5 Hz at -14 dBm + 2 dB, cycling ON for 2 seconds ± .4 seconds and OFF for 4 seconds ± .8

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seconds, shall be sent continuously on the send terminals. With the insertion of the operator's answer cord into the channel's subscriber jack, the ring back shall stop, the night alarm deactivated and service request indicator reset. Communication between the calling and called parties shall then be possible.

3.8.1.2 Outgoing Calls. Each channel of the equipment shall extend calls to a 4-wire automatic switchboard. Upon operation of the seize and access switch, a seize signal of 2250 Hz ± 29 Hz at -14 dBm ± 2 dB shall appear on the send terminals, a dial tone of 425 Hz at -14 dBm received by the equipment shall be audible to the operator. Depressing access switch and each keysender digit shall send the appropriate dual tone (See Table V) ± 1.3% at -14 dBm ± 2 dB on the send terminal.

3.8.1.3 Release.

3.8.1.3.1 Release Transmission. Upon operation of the release and access switch on the equipment, a release signal of 2600 Hz ± 33.8 Hz at -14 dBm ±2 dB shall appear on the send terminals. When a release acknowledge signal of 570 Hz appears on the receive terminals it shall be detected and the 2600 Hz release signal shall cease between 3 and 10 seconds later.

3.8.1.3.2 Release Detection. Receipt of a release signal of 2600 Hz shall cause the equipment to return a release acknowledge signal of 570 Hz ± 7.4 Hz at -14 dBm ± 2 dB on the send terminals, the service request indicator and night alarm shall be activated. The release acknowledge signal shall cease 250 ms after release signal ceases.

3.8.1.4 Audible Alarm. An audible alarm, capable of being adjustable for tone level, shall operate upon detection of seize or release on any channel until reset.

3.8.1.5 Tone Detectors. The equipment shall respond to received tones ± 1.5% of the nominal frequency (See Table VI) and signal power of -5 dBm and -30 dBm, and any level between, at the converter input. The seize detector shall not respond to a release tone of 2561 Hz at -5 dBm. The release detector shall not respond to a seize tone of 2284 Hz at -5 dBm.

3.8.2 Electrical Transmission Characteristics.

3.8.2.1 Return loss. When referenced to a 600 ohm resistive load, the return loss shall be equal to or greater than the value specified in the following table:

Frequency (Hz)	Return Loss (dB)
300-500	14
501-800	17
801-2800	20
2801-3200	17
3201-3500	14

3.8.2.2 Transmission loss. The transmission loss of an established connection shall not exceed 5 dB over the frequency range of 300 to 3500 Hz.

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3.8.2.3 20 Hz rejection. The 20 Hz transmission loss from a 2-wire to 4-wire connection shall be greater than 11.2 dB when a signal of 90 volts rms is applied to the 2-wire terminations.

3.8.2.4 Longitudinal balance. The longitudinal balance of all terminal pairs shall be at least 40 dB over the frequency range of 0.30 KHz to 3.5 KHz.

3.8.2.5 Harmonic distortion. Any harmonic which may be produced by transmission through the equipment shall be at least 35 dB below the fundamental within the frequency range of 0.30 KHz to 3.5 KHz transmitted at zero dBm level.

3.8.2.6 Hybrid transhybrid loss. The hybrid transhybrid loss, from 4-wire receive to 4-wire transmit, shall exceed 30 dB when the 2-wire output is terminated in a 600 ohm ± 1% resistor.

3.8.2.7 Crosstalk. The crosstalk coupling loss between channels shall be at least 50 dB when referred to equal level points.

3.8.2.8 Frequency response. On any connection through the equipment, the frequency attenuation characteristic shall be flat within ± 1 dB with respect to the 1000 Hz level over the range 0.5 KHz to 3.5 KHz. From 0.3 to 0.5 the response shall be between +1 dB and -3 dB with respect to the 1000 Hz level. Above 3.5 KHz, the response shall not exceed +1 dB above the 1000 Hz level.

3.8.2.9 Noise level. The noise level of any voice transmission circuit within the equipment shall not exceed 23 dBa when measured with Transmission Measuring Set TS-559/FT, or equal, with F1A weighting.

3.8.3 Power supply. The equipment shall operate from four batteries, BA-30 or equivalent, with no degradation from the requirements of this specification between the limits of 1.6 volts to 1.1 volts per cell. In the all channels idle condition, the equipment shall have a maximum current drain of 20 ma at 6.4 volts. In the all channels busy condition, the equipment shall have a maximum current drain of 30 ma at 6.4 volts.

3.8.4 Operational Inspection. The equipment shall perform its intended function when subjected to the test of 4.8.4.

3.9 Service Conditions. The equipment shall meet the following service conditions and provide full specification performance during and after testing.

3.9.1 Temperature.

3.9.1.1 High. When tested as specified in 4.9.1.1, the equipment shall meet the requirements of 3.8.1 through 3.8.4.

3.9.1.2 Low. When tested as specified in 4.9.1.2, the equipment shall meet the requirements of 3.8.1 through 3.8.4.

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3.9.2 Humidity. The equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the requirements of 3.8.1 through 3.8.4 during the operating times specified in 4.9.2.

3.9.3 Altitude. The equipment shall operate at altitudes up to 10,000 feet above sea level after storage of 50,000 feet and meet the requirements of 3.8.1 through 3.8.4 when tested as specified in 4.9.3.

3.9.4 Leakage (Immersion). The equipment in its field transport mode shall be capable of being immersed to a covering depth of 3 feet of fresh water for a period of 2 hours without evidence of any water leakage. (See 4.9.4.).

3.9.5 Salt Fog. After the salt fog test of 4.9.5, the equipment, when examined visually with the aid of a 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface, or exfoliation of metal. There shall be no clogging or binding of controls or other moving parts after subjecting the equipment to the 48 hour salt fog exposure period or after the 48-hour drying period.

3.9.6 Fungus. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.9.6. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.9.7 Air Seal. When the equipment is tested as specified in 4.9.7, the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge (psig), per minute.

3.9.8 Bounce, Loose Cargo. The equipment shall sustain no damage after being subjected to the vibration and shock of 4.9.8. Cosmetic damage such as scratches and minor dents will be allowed. The equipment shall meet the requirements of 3.8.1 through 3.8.4 following the test.

3.9.9 Shock, Drop. When tested in accordance with 4.9.9, the equipment shall meet the requirements of 3.8.1 through 3.8.4. The equipment shall sustain only minor physical damage.

3.9.10 Bench Handling. After testing as specified in 4.9.10, the equipment shall meet the requirements of 3.8.1 through 3.8.4. The equipment shall sustain no damage.

3.9.11 Sand and Dust. When tested in accordance with 4.9.11, the equipment shall meet the requirements of 3.8.1 through 3.8.4.

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3.10 Fabric and Thread. Fabric and thread shall be preshrunk in order to provide for satisfactory fit of finished items. The canvas bag and strap, after soaking in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature shall not exhibit any noticeable shrinkage (See 4.12).

3.11 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268.

3.12 Electromagnetic Compatibility (See 4.11).

3.12.1 Electromagnetic emissions in the frequency range of 30 Hz to 20KHz shall not appear on signal leads in excess of the values shown in Figure 1, 2, 3, and 4.

3.12.2 Radiated electric field emissions in the frequency range of 14 KHz 50 400 MHz shall not be high than the values shown in Figure 5 and 6.

3.12.3 The required susceptibility field strength is one volt/meter over the frequency range of 14 KHz to 1 GHz. The converter shall not lose any calls once the call has been made or indicate a change in status when the case and cables are subjected to this field intensity.

3.13 Reliability. The equipment shall have a specified mean-time-between-failure (θ_o) of 1000 hours when operated under the following conditions:

(1) Ambient temperature: 50°C

(2) Ground environment

(3) Continuous operation

The testing as specified in 4.13 herein will be conducted to determine compliance to the reliability requirement above.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection. (See 4.3). Does not include preparation for delivery.

b. Inspections covered by subsidiary documents (See 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of preparation for delivery (See 4.14).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish six (6) First Article units of the complete Converter, Telephone Signal CV-1919( )/G.

4.3.2 First Article Inspection. The First Article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B, Group C (3 units) and Group D (3 units) (See Tables II, III, IV, and V respectively). The inspection shall be performed in the following order: 4.4, 4.6.1, 4.6.2, Group A and Group B for all units; Group C for three units as specified in Table I and Group D for three units. After completion of Group C or Group D tests, conforming units shall be reinspected and shall pass Group A and Group B inspection.

4.3.3 First Article Data. The First Article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE I

Order of First Article and Group C Tests

Inspection	Number of Units					
	If one	If two		If three		
		Unit 1	Unit 2	Unit 1	Unit 2	Unit 3
Temperature						
Low	1	1		1		
High	2	2		2		
Altitude	3		1		1	
Humidity	4	3		3		
Fungus	5		5			4
Salt Fog	6	4			2	
Bounce, loose cargo	7		2			1
Shock, bench handling	8		3			2
Shock, drop	9		4			3

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4.4 Inspections Covered by Subsidiary Documents: The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Construction	3.2
Parts, materials and Processes-general	3.3
Finish, protective	3.4
Marking	3.5

4.5 Quality Conformance Inspection of Equipment Before Preparation For Delivery. The contractor shall perform the inspection specified in 4.4, 4.6 and 4.5.1 thru 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The inspection of 4.6.2 shall be performed subsequent to 4.4 and 4.6.1 and prior to 4.5.1.

4.5.1 Group A Inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II, Discrete lots shall be formed from units that pass this inspection. Fastors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of Inspection Within Group A. Group A inspection shall be performed in the order listed in Table II.

TABLE II
Group A Inspection

Inspection	Reqmt Para	Test Para	AQL Major	AQL Minor
Visual/Mechanical	3.11	4.10	1%	4%
Shrinkage	3.10	4.12	1%	
Operation				
Incoming Calls	3.8.1.1	4.8.1.1		
Outgoing Calls	3.8.1.2	4.8.1.2	1.0 DPHU for	
Release	3.8.1.3	4.8.1.3	the group	
Audible Alarm	3.8.1.4	4.8.1.4		
Tone Detectors	3.8.1.5	4.8.1.5		
Operation Inspection	3.8.4	4.8.4	1%	
Air Seal	3.9.7	4.9.7	1%	

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NOTE: *DPHU - Defects Per Hundred Units:* Defects in all characteristics other than Visual and Mechanical are considered major.

4.5.2 Group B Inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III.

4.5.2.1 Group B Sampling Plans. The Group B sampling plans, for the AQL's listed in Table III, shall be inspection level S-4 of MIL-STD-105.

4.5.2.2 Order of Inspection Within Group B. Group B inspection shall be performed in the order listed in Table III.

TABLE III
Group B Inspection

Inspection	Reqmt Para	Test Para	AQL Major
Interchangeability	3.7	4.7	6.5%
Transmission	3.8.2	4.8.2	4.0 DPHU for the group
Power	3.8.3	4.8.3	
Immersion	3,9.4	4.9.4	6.5%

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V . Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Samples for Group C Inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every 120 units or fraction thereof.

4.5.3.1.2 Subgroup 2. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, 1 unit shall be selected every 240 units or fraction thereof.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, 1 unit shall be selected every 500 units or fraction thereof.

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TABLE IV
Group C Inspection

Inspection	Reqmt Para	Test Para
Subgroup 1		
Temperature	3.9.1	4.9.1
Humidity	3.9.2	4.9.2
Subgroup 2		
Altitude	3.9.3	4.9.3
Bounce, Loose Cargo	3.9.8	4.9.8
Shock, Drop	3.9.9	4.9.9
Shock, Bench Handling	3.9.10	4.9.10
Electromagnetic Compatibility	3.9.12	4.11
Subgroup 3		
Salt Fog	3.9.5	4.9.5
Fungus	3.9.6	4.9.6
Sand and Dust	3.9.11	4.9.11

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in the order shown in Table I.

4.5.3.3 Group C Failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Group D Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

4.5.4.1 Sampling for Group D Inspection.

4.5.4.1.1 First Article Sampling. Three (3) samples shall be selected at random for inspection in accordance with Table V.

4.5.4.1.2 Production Sampling. Four (4) units shall be selected at random and subjected to the test of Table V each two months of production. Selection of units from the second month of each two month period may be selected early in the second month so that testing may commence early in the second month and be completed by the end of the second month. Group D inspection shall be in accordance with Table V.

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TABLE V
Group D Inspection

Inspection	Reqmt Para	Test Para
Reliability	3.13	4.13

4.5.5 Reinspection of Conforming Group C and Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C or Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A and Group B inspection.

4.6 Preconditioning.

4.6.1 Bounce Preconditioning. The equipment shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection.

4.6.2 Burn-In. Each equipment on order shall be subjected to a 48 hour (on-time) debugging (burn-in) period with the last 24 hours failure free (on-time) (3.6).

4.6.2.1 Test Conditions. The test conditions for this test shall be the same as for reliability (see 6.8 of SM-A-743617 and 4.13.3 of this specification).

4.6.2.2 Monitoring. The operational test procedure of 6.8 of SM-A-743617 (6.8.2) shall be performed at least once at 24 hours of "on-time" and at the end of the 48 hour ("On-time" minimum test time) to determine if a failure has occurred in burn-in. If additional testing is required to satisfy the 24 hour failure free requirement of 3.6, it shall be accomplished.

4.6.2.3 Failure. A failure is defined as any deviation in performance as noted from that specified in 6.8.2 of SM-A-743617.

4.6.2.4 Treatment of Failures. If a failure occurs during the first 24 hours of burn-in, it shall be corrected and returned to complete the burn-in as required in 3.6. If a failure occurs during the last 24 hours of burn-in, it shall be corrected and resubjected to burn-in until it successfully completes 24 hours failure free.

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4.7 Inspection for Dimensional Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect. Additionally, the units shall be subjected to the Interchangeability Test of paragraph 6.9 of SM-A-743617.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions or cavaties, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies or major unit.

e. Size and form of special threads.

4.8 Electrical Performance.

4.8.1 Operation.

4.8.1.1 Incoming Call. The equipment shall be subjected to the Incoming Call Test of SM-A-743617, para 6.3 and meet the requirements of 3.8.1.1.

4.8.1.2 Outgoing Call. The equipment shall be subjected to the Outgoing Call Test of SM-A-743617, para 6.4 and meet the requirements of 3.8.1.2.

4.8.1.3 Release. The equipment shall be subjected to the Release Test of SM-A-743617, para 6.1 and meet the requirements of 3.8.1.3.

4.8.1.4 Audible Alarm. The equipment shall be subjected to the Audible Alarm Test of SM-A-743617, para 6.2 and meet the requirements of 3.8.1.4.

4.8.1.5 Tone Detectors. The equipment shall be subjected to para 6.5 of SM-A-743617 and meet the requirements of 3.8.1.5.

4.8.2 Transmission.

4.8.1.2 Return Loss. The equipment shall be subjected to the Return Loss Test specified in para 6.11 of SM-A-743617, and meet the requirements of 3.8.2.1

4.8.2.2 Transmission Loss. The equipment shall be subjected to the Transmission Loss Test specified in para 6.12 of SM-A-743617 and meet the requirements of 3.8.2.2.

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4.8.2.3 20 Hz Rejection. The equipment shall be subjected to the 20 Hz Rejection Loss Test described in para 6.18 of SM-A-743617 and meet the requirements of 3.8.2.3.

4.8.2.4 Longitudinal Balance. The equipment shall be subjected to the Longitudinal Balance Test described in para 6.14 of SM-A-743617 and meet the requirements of 3.8.2.4.

4.8.2.5 Harmonic Distortion. The equipment shall be subjected to the Harmonic Distortion Test described in para 6.16 of SM-A-743617 and meet the requirements of 3.8.2.5.

4.8.2.6 Hybrid Transhybrid Loss. The equipment shall be subjected to the Hybrid Transhybrid Loss Test described in para 6.13 of SM-A-743617 and meet the requirements of 3.8.2.6.

4.8.2.7 Cross Talk. The equipment shall be subjected to the Cross Talk Test described in para 6.15 of SM-A-743617 and meet the requirements of 3.8.2.7.

4.8.2.8 Frequency Response. The equipment shall be subjected to the Transmission Loss Test described in para 6.12 of SM-A-743617 and meet the requirements of 3.8.2.8.

4.8.2.9 Noise Level. The equipment shall be subjected to the Noise Level Test described in para 6.17 of SM-A-743617 and meet the requirements of 3.8.2.9.

4.8.3 Power Supply. The equipment shall be subjected to the Power Supply Test described in para 6.10 of SM-A-743617 and meet the requirements of 3.8.3.

4.8.4 Operational Inspection. The equipment shall be subjected to the Operational Inspection Test described in para 6.7 of SM-A-743617 and meet the requirements of 3.8.4.

4.9 Service Conditional Tests.

4.9.1 Temperature Test.

4.9.1.1 High Temperature. The equipment shall be subjected to the test of Method 501.1,Procedure II, of MIL-STD-810 and meet the requirements of 3.9.1.1. The storage temperature shall be 160°F and high operating temperature shall be 150°F.

4.9.1.2 Low Temperature. The equipment shall be subjected to the test of Method 502.1,Procedure I, of MIL-STD-810 and meet the requirements of 3.9.1.2. The storage temperature shall be -70°F with storage of 2 hours after equipment's temperature has stabilized. The low operational temperature shall be -25°F.

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4.9.2 Humidity Test. The equipment shall be subjected to the test of Method 507.1,Procedure III, of MIL-STD-810 and meet the requirements of 3.9.2. Measurements shall be made during the last 5 hours of each 48 hour cycle, and at the end of each 240 hours in the 480 hour soak period.

4.9.3 Altitude Test. The equipment shall be subjected to the test of Method 500.1,Procedure I, of MIL-STD-810 and meet the requirements of 3.9.3.

4.9.4 Immersion Test. The equipment shall be subjected to the test of Method 512.1,Procedure I, of MIL-STD-810 and meet the requirements of 3.9.4.

4.9.5 Salt Fog. The equipment shall be subjected to the test of Method 509.1, Procedure I, of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any flakey, pitted, blistered, or otherwise loosened finish or metal surface, and meet the requirements of 3.9.5.

4.9.6 Fungus. The equipment shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermatic sealed test items shall be opened during exposure and internal areas shall be sprayed with spore suspension as well as external surfaces. Operation of the equipment is not required but examination at the conclusion of the exposure period shall reveal no evidence of viable fungus on any surface.

4.9.7 Air Seal. Remove the two-way pressure relief valve from the case and install a test valve. The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported, shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case. The vacuum then shall be valved-off and the interior pressure measured during the ensueing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

4.9.8 Bounce, Loose Cargo. The equipment shall be tested in accordance with Method 514.2,Procedure XI of MIL-STD-810, and meet the requirements of 3.9.8.

4.9.9 Shock,Test, Drop. The equipment shall be tested in accordance with Method 516.2,Procedure II of MIL-STD-810, and meet the requirements of 3.9.9.

4.9.10 Bench Handling. The equipment shall be tested in accordance with Method 516.2,Procedure V, of MIL-STD-810, and meet the requirements of 3.9.10.

4.9.11 Sand and Dust. The equipment shall be tested in accordance with Method 510.1, Procedure I, MIL-STD-810, and meet the requirements of 3.9.11 both before and after testing. During testing the equipment will be in the operating mode but not operating.

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4.10 Visual and Mechanical Inspection. The equipment shall be examined for defects listed in MIL-STD-252. The equipment shall also be inspected for conformance with MIL-STD-454, Requirement 9 and the applicable portions of MIL-P-11268.

4.11 Electromagnetic Compatibility.

4.11.1 First Article Tests. To determine compliance with 3.12 during First Article Testing, the equipment shall be tested in accordance with paragraph 6.6 of SM-A-743617.

4.11.2 Production Tests. To determine compliance with 3.12 during production testing, the equipment shall be tested in accordance with paragraph 6.6 of SM-A-743617 except that 6.6.4 and 6.6.5 shall be omitted.

4.12 Test for Shrinkage. The Canvas Bag and Strap shall be inspected for conformance to 3.10.

4.13 Reliability Testing.

4.13.1 Reliability Qualification Test. To determine compliance with 3.13 during First Article testing, the equipment shall be tested in accordance with test plan IV of MIL-STD-781 and paragraph 6.8 of SM-A-743617.

4.13.2 Reliability Production Test. To determine compliance with 3.13 during production, the equipment shall be tested in accordance with test plan IV A, MIL-STD-781 and paragraph 6.8 of SM-A-743617.

4.13.3 Reliability Test Conditions. The test described in 4.13.1 and 4.13.2 above shall be conducted using test level C of MIL-STD-781 except that the "on-off" cycling shall be as follows: The equipment will be continuously on with one instantaneous "on-off" cycle for each 3 hours of "on-time". The following paragraphs of MIL-STD-781 shall apply:

Failure Action (5.5)

Verifying Repair (5.6)

Restoration of Failed Equipment (5.9)

Test Records (5.10)

4.13.4 Failure. A failure is defined as any deviation in performance as noted from that specified in 6.8.2 of SM-A-743617.

4.14 Quality Conformance Inspection of Preparation for Delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5. If testing shall be required, the Operation Inspection Test of 4.8.4 shall be performed.

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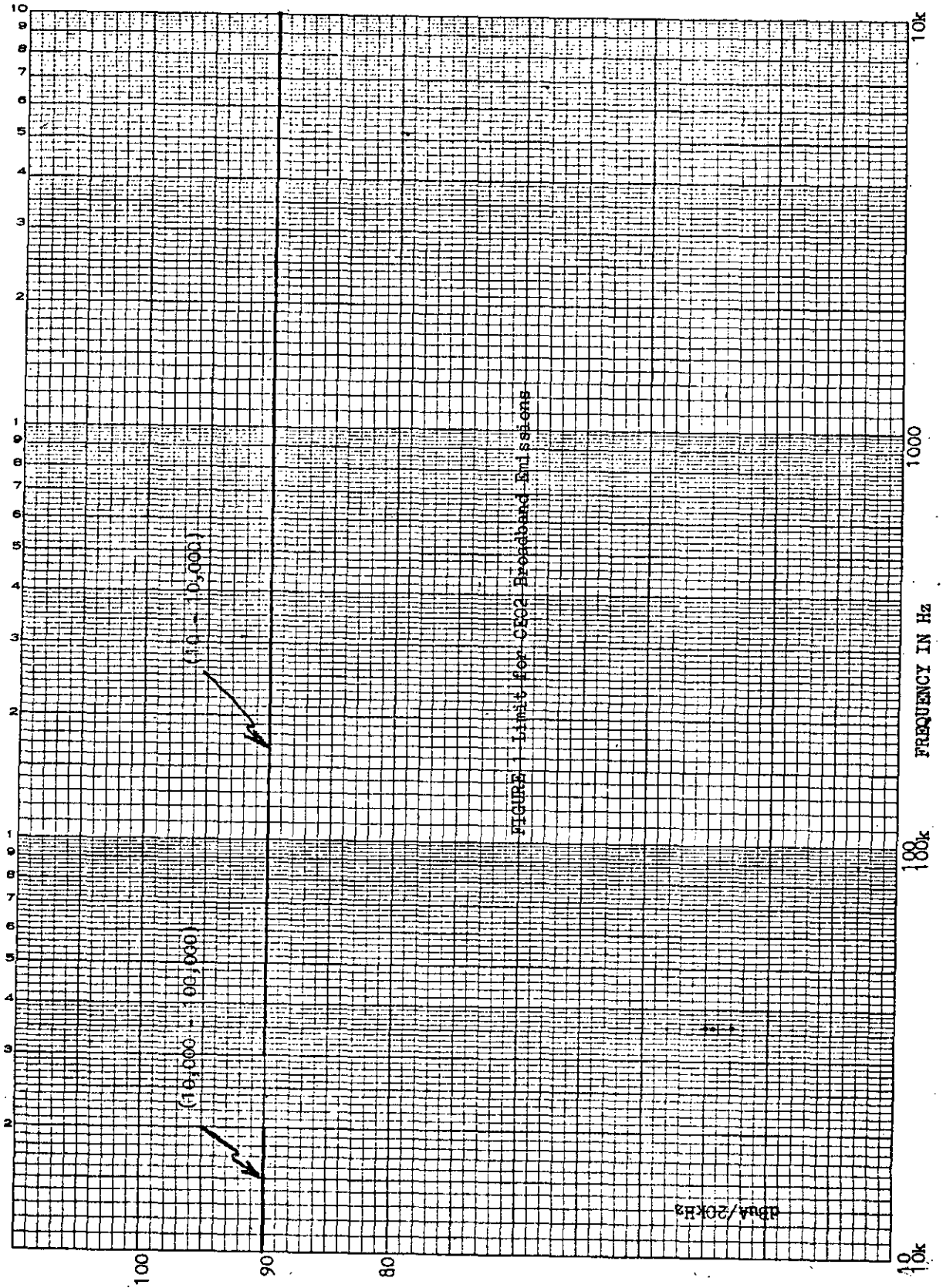


FIGURE 1. Limit for CE02 Broadband Emissions

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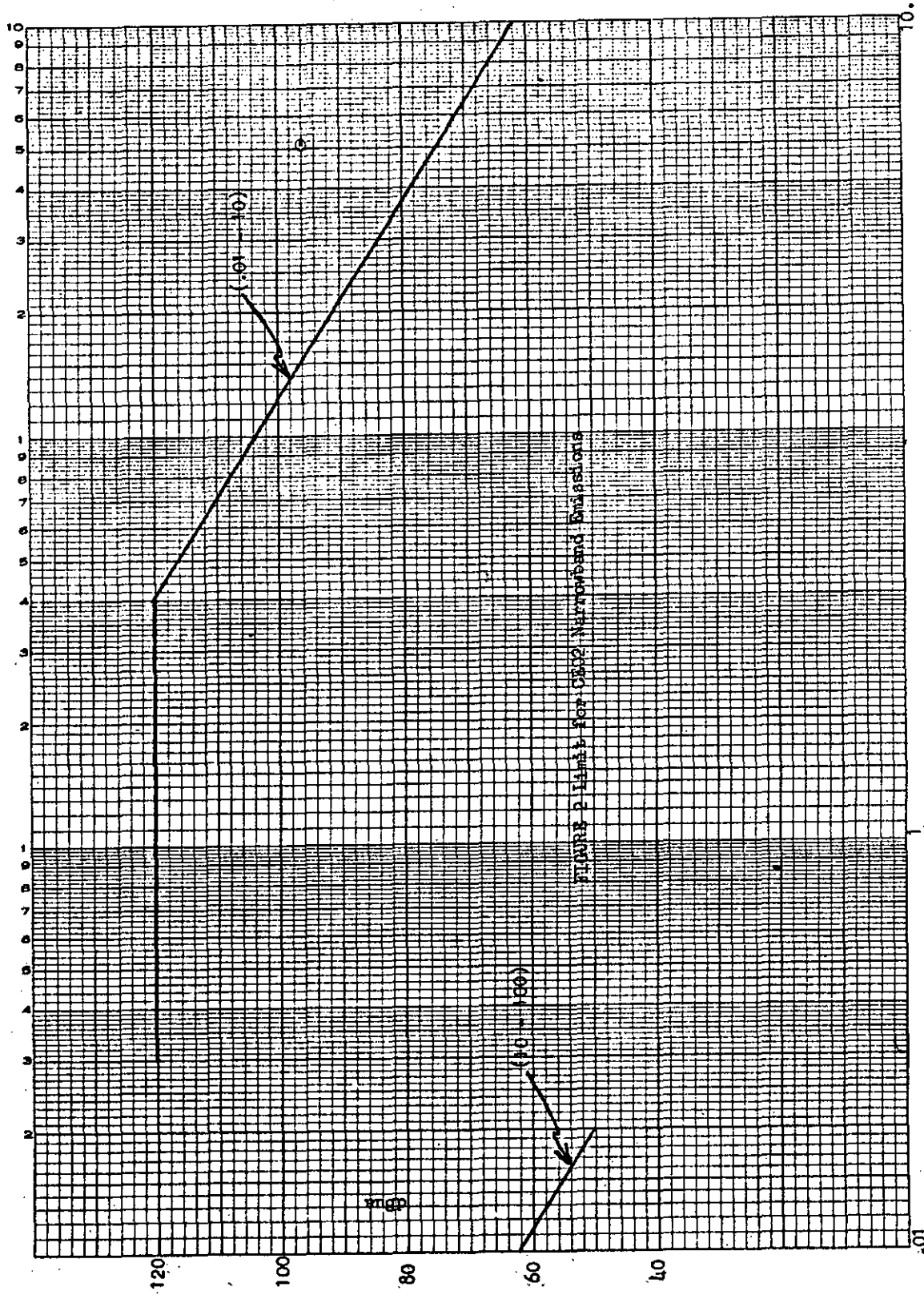


FIGURE 2. Limit for CS02 Narrowband Emissions

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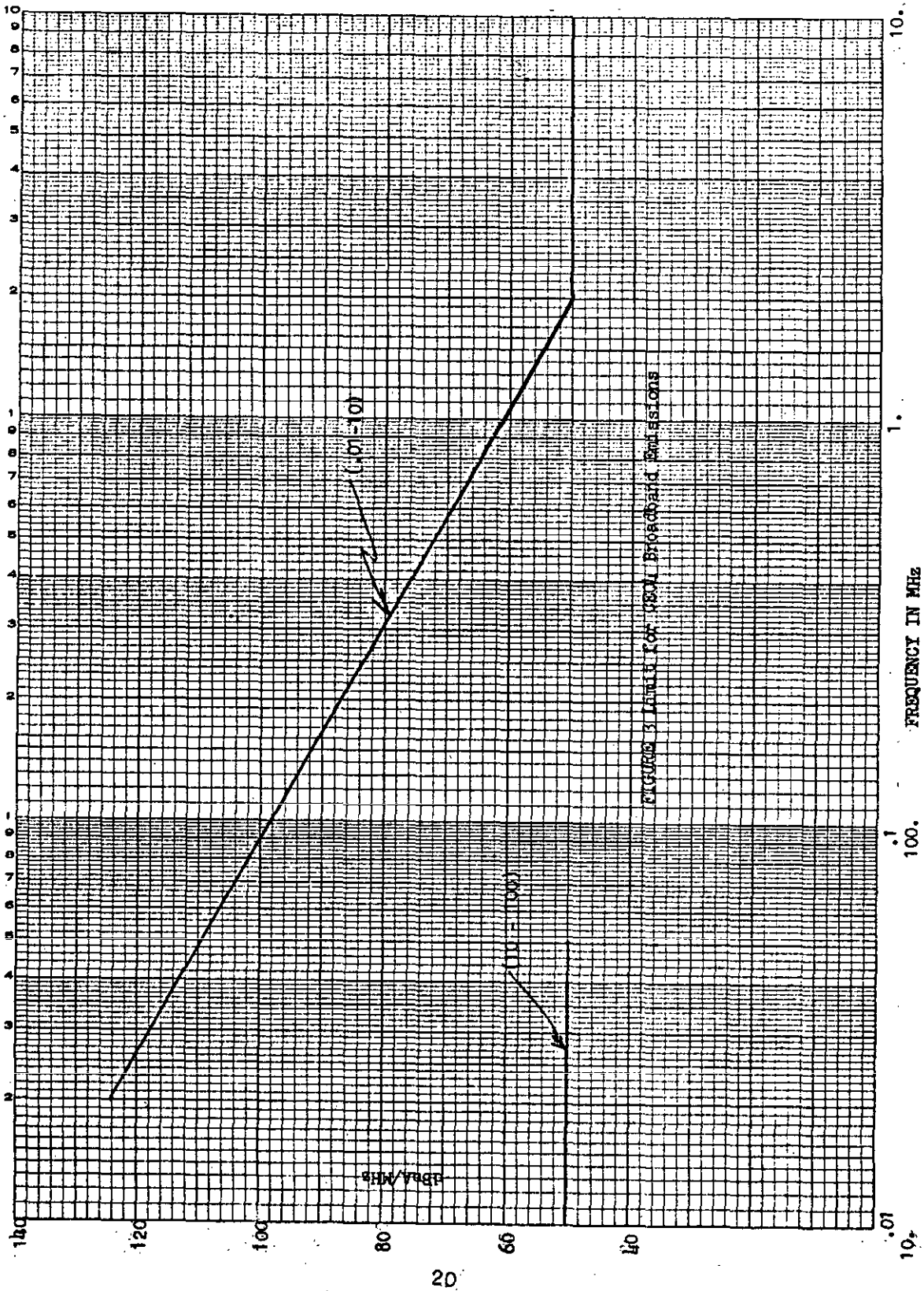
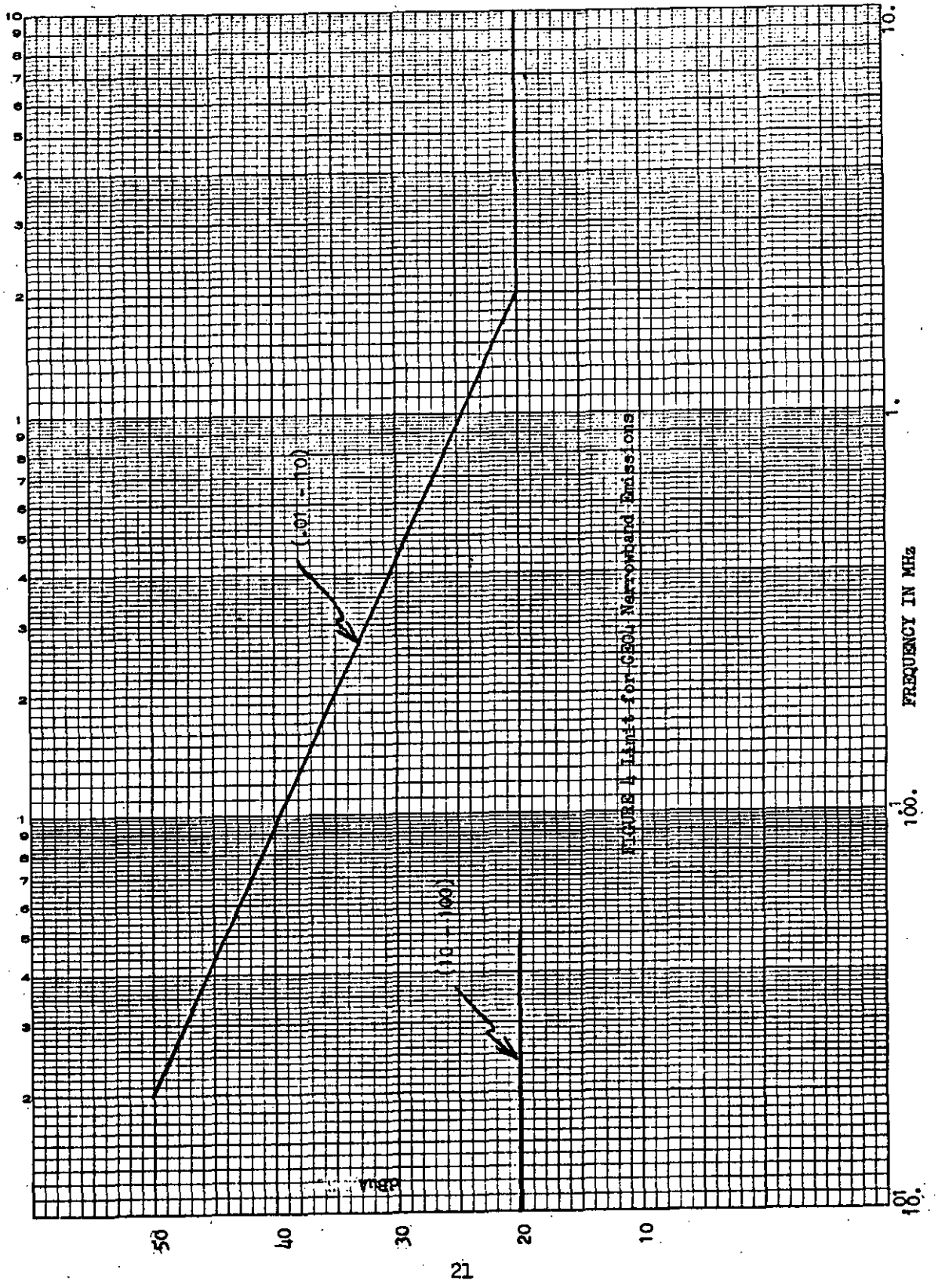


FIGURE 3. Limits for CE01 Broadband Emissions

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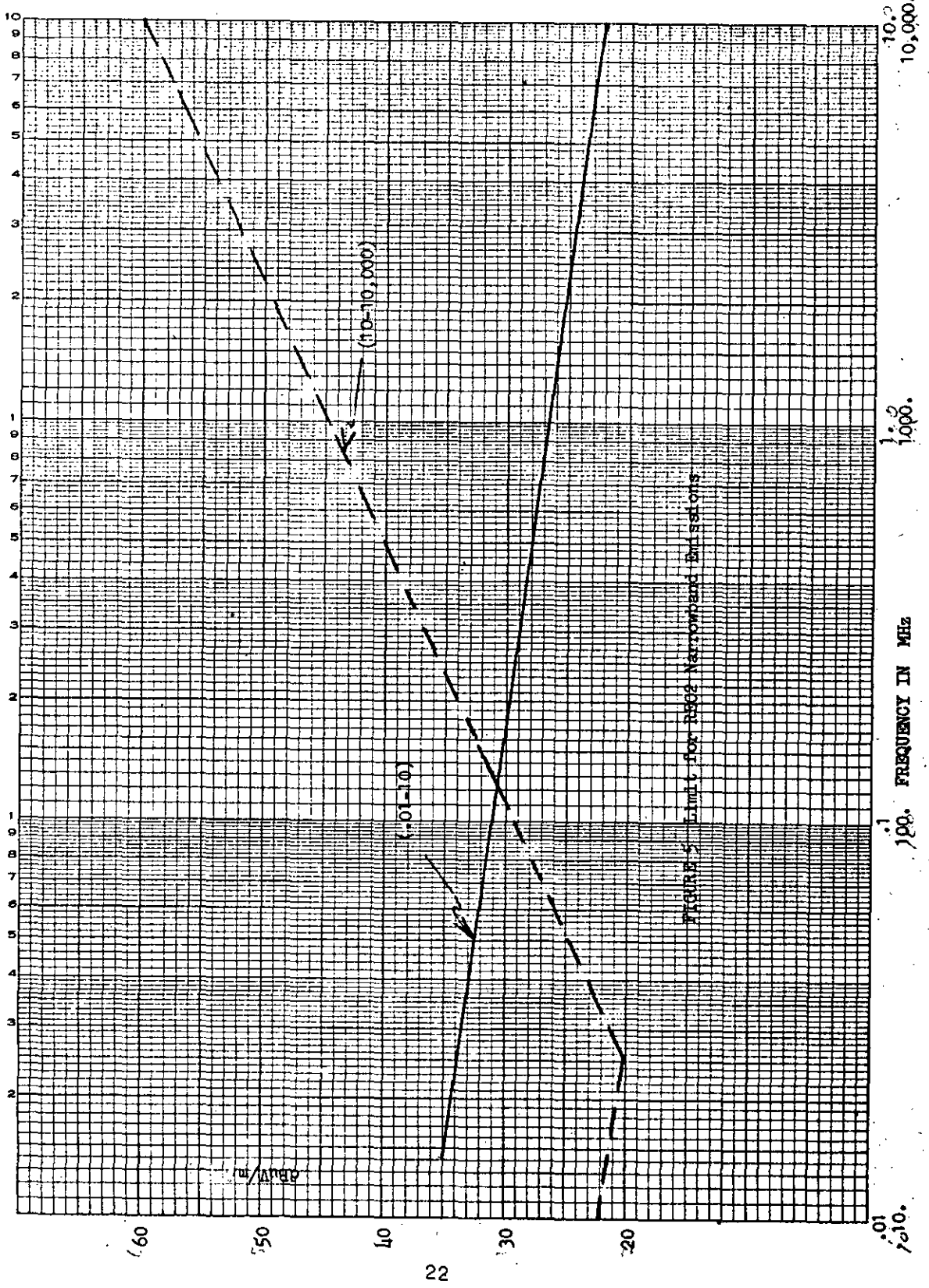


FIGURE 5. Limit for RS02 Narrowband Emissions

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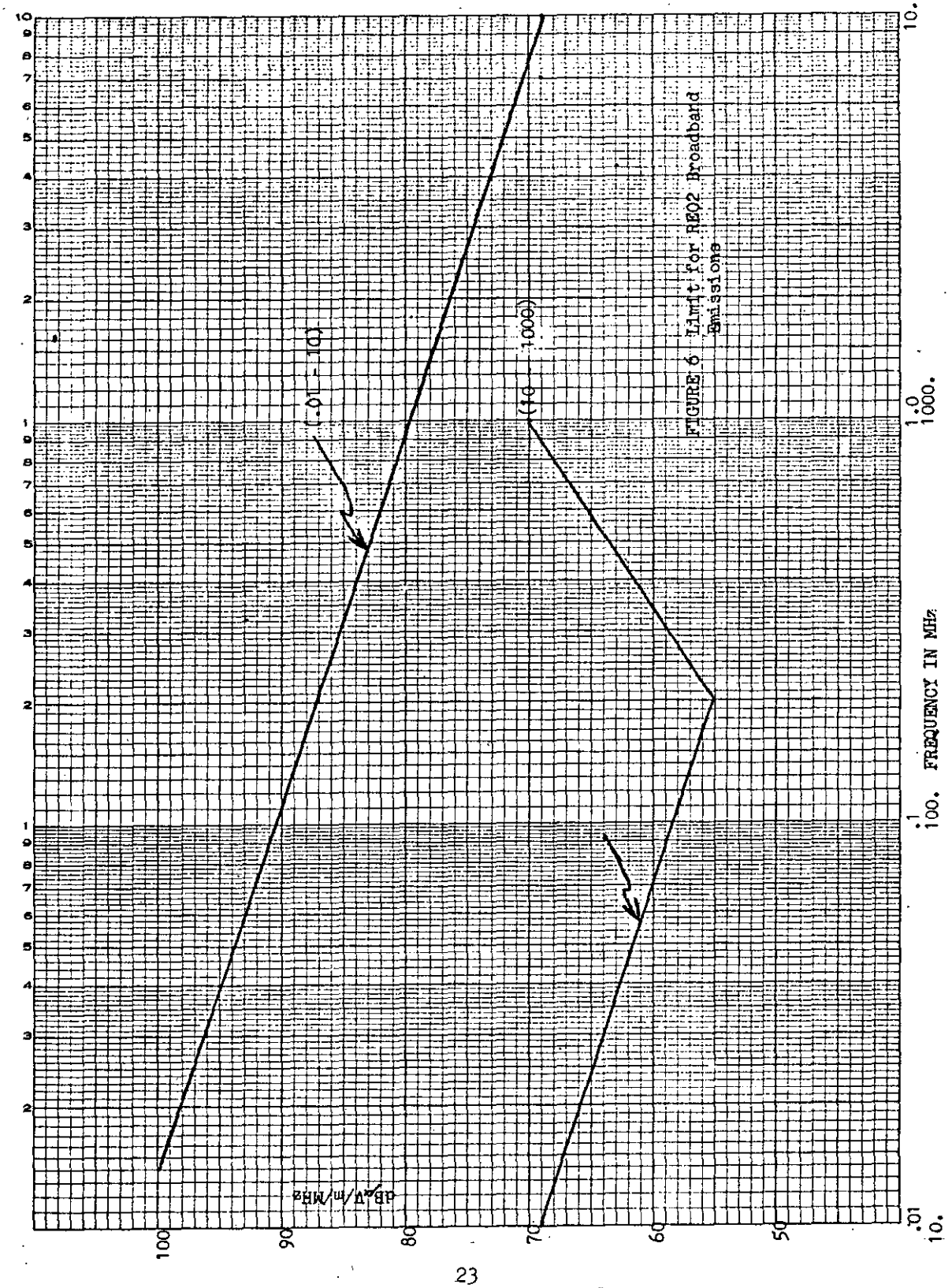


FIGURE 6. Limit for RE02 Broadband Emissions

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TABLE VI

CONVERTER GENERATED SIGNALS

Function	Freq. in Hz	Single Tone Level in dBm	Duration	Remarks
Digits 1 2 3 4 5 6 7 8 9 0	697/1209 697/1336 697/1477 770/1209 770/1336 770/1477 852/1209 852/1336 852/1477 941/1366	-14 ± 2	Operator Controlled	Operator originated
Seize	2250	-14 ± 2	Operator controlled	Held until seize ack. is heard
Seize acknowledge	570	-14 ± 2	Continuous until 250 ms after seize stops	
Ringback tone	425	-14 ± 2	Until operator answers	2 s on 4 s off
Conference	941/1477	-14 ± 2	Operator controlled	Operator originated
Recall	941/1209	-14 ± 2	Operator controlled	Operator originated
Release	2600	-14 ± 2	Continuous up to 3 to 10 s	Operator originated. Shutoff by release ack.
Release acknowledge	570	-14 ± 2		Cutoff 250 ms after release ends

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Converter, Telephone Signal CV-1919( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Converter, Telephone Signal CV-1919( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Converter, Telephone Signal CV-1919( )/G. Each converter shall be individually packaged method III as follows: Place mounting strap within canvas bag and store in cover of the converter. Secure the cover on the converter with the fasteners provided. Cushion each converter on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, variety SW, Grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-26514. Place each item within a close-fitting box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. Converter, Telephone Signal CV-1919( )/G shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2).

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5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Converter, Telephone Signal CV-1919( )/G, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A or shrink film.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of Converter, Telephone Signal CV-1919 ( )/G, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of items, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

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5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of Converter, Telephone Signal CV-1919 ( )/G, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of items, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively; and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES.

6.1 Intended Use. The equipment is intended to provide the necessary conversion to interface manual 2-wire SB-22 and SB-86 switchboards with 4-wire Central Office, Telephone Automatic AN/TTC-38( ) (V) switchboards.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, date of this specification and any amendment thereto.

b. Level of packaging, packing, and preservation (Level A, Level B, or Level C.

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. When Rough Handling test is required.

6.3 Group C Inspection. Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

Custodian:

Army (EL)

Preparing Activity

Army (EL)

Project Number 5805-A202

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-49073 CONVERTER, TELEPHONE SIGNAL CV-1919( )/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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